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CBA-Whisper: Curriculum Learning-Based AdaLoRA Fine-Tuning on Whisper for Low-Resource Dysarthric Speech Recognition

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Literature Review

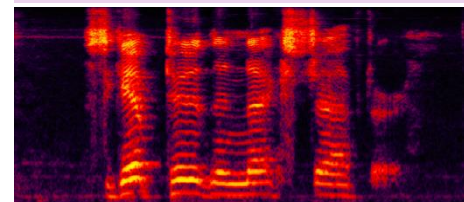


Review– 1.1 Introduction

➤ What is dysarthric speech?

- a representative type of low-resource speech
- a motor speech disorder caused by conditions such as Parkinson's disease (PD), multiple sclerosis (MS), and other neurological impairments
- the unique features of dysarthric speech, such as **poor articulation**, **decreased volume and clarity**, **slower speaking rates** and **increased dysfluencies**, not only pose significant challenges for ASR systems, but also heighten the likelihood of machine hallucinations

Skip to the next chapter.



➤ What is Speech Accessibility Project Challenge (SAPC)?

- The goal of the SAPC is to rapidly advance the state of the art in dysarthric speech recognition.
- Competitors can use data from the Speech Accessibility Project 2024-04-30 dataset (train: 290 hours from 369 speakers, dev: 44 hours from 59 speakers) to train an ASR.
- The team with the lowest WER on Test2, and the team with the highest semantic score on Test2, will each receive a prize of US \$5000.

Review– 1.2 Previous Approaches (1/3)

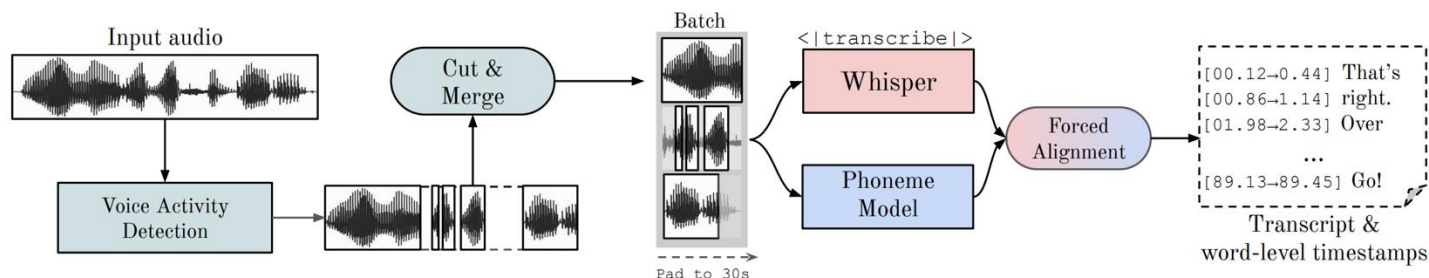
➤ Previous ASR Models

- **Canary-1b**, but the results weren't great [18.27, 76.80]
- **CrisperWhisper** has an issue with slow timestamps
- **Whisper large-v3** series is difficult to fine-tune [11.66, 84.04]
- A viable option is to choose **Whisper large-v2** [12.18, 82.50]

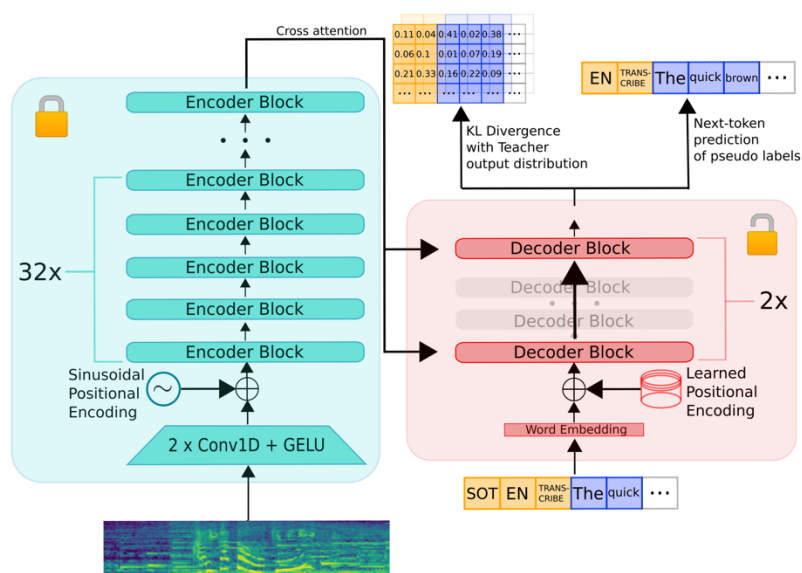
model ▲	Average WER ▼ ▲	RTFx ▲ ▲	AMI ▲	Earnings22 ▲	Gigaspeech ▲
nvidia/canary-1b	6.5	235.34	13.9	12.19	10.12
nyrahealth/CrisperWhisper	6.67	84.05	8.71	12.89	10.24
nvidia/parakeet-tdt-1.1b	7.01	2390.61	15.87	14.49	9.52
nvidia/parakeet-rnnt-1.1b	7.12	2053.15	17.01	13.94	9.89
nvidia/parakeet-ctc-1.1b	7.4	2728.52	15.67	13.75	10.28
openai/whisper-large-v3	7.44	145.51	15.95	11.29	10.02
nvidia/parakeet-tdt_ctc-110m	7.49	5345.14	15.89	12.37	10.52
nvidia/parakeet-rnnt-0.6b	7.5	2815.72	17.4	14.66	10.01
distil-whisper/distil-large-v3	7.52	214.42	15.16	11.79	10.08
nvidia/parakeet-ctc-0.6b	7.69	4281.53	16.46	14.26	10.39
openai/whisper-large-v3-turbo	7.83	200.19	16.13	11.63	10.14
openai/whisper-large-v2	7.83	144.45	16.74	12.05	10.67

Review– 1.2 Previous Approaches (2/3)

➤ Previous Whisper-based Approaches



Bain, Max et al. "WhisperX: Time-accurate speech transcription of long-form audio" INTERSPEECH. 2023.



Sanchit Gandhi et al. "Distil-Whisper: Robust Knowledge Distillation via Large-Scale Pseudo Labelling" arXiv. 2023.

- **Faster-Whisper** (based on CTranslate2) focus on improving inference speed.
- **Whisper Turbo** (pruned, finetuned) and **Distil-Whisper** (pruned, distil) focuses on model compression.
- **WhisperX** emphasizes enhancing model performance.

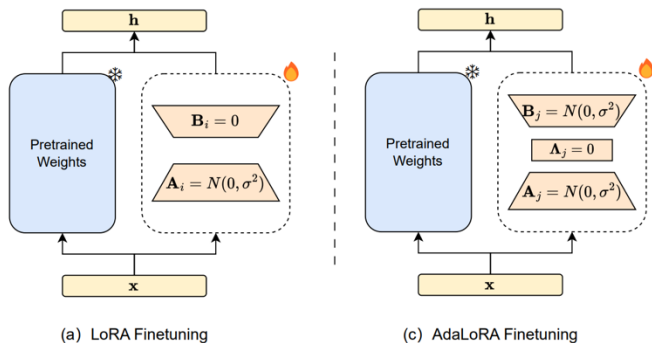
Tab. 1 4 Whisper variants implementation and comparison.

Model	Time [s]	Transcription review
Whisper turbo	55	Good, but sometimes hallucinated.
Faster-Whisper	3.3	Almost Good
WhisperX	29.9	Excellent
Distil-Whisper	22.7	Almost Good

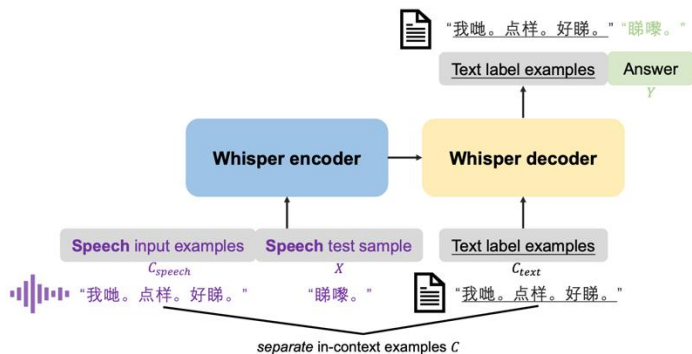
Review– 1.2 Previous Approaches (3/3)

➤ Previous Approaches for Whisper Fine-Tuning

- AdaLoRA > LoRA > prompt-ft > full-ft > original large-v2



Xu, Tianyi et al. "Towards Rehearsal-Free Multilingual ASR: A LoRA-based Case Study on Whisper" INTERSPEECH. 2024.



Wang, Siyin et al. "Can whisper perform speech-based in-context learning?" ICASSP. 2024.

Table 3: Performance (CER%) of Whisper large-v2 finetuned on Chinese dataset.

	original large-v2	full-ft	LoRA	AdaLoRA
Aishell-1	8.81	5.72	4.23	3.78
In house Chinese	47.94	40.57	36.98	34.43
#Trainable Params	-	100%	3.47%	1.36%

Table 3. %WERs of spontaneous speech using the iFLYTEK test set. k is the number of in-context examples, selected by k NN with the example retrieval model $\Theta = L$. The example presentation order is "far to near". The example datastore is the RASC863 dialectal word dataset of corresponding dialect.

Setting	Corpus & Whisper model size							
	iFLYTEK Chongqing				iFLYTEK Guangzhou			
	B	S	M	L	B	S	M	L
LID	75.9	68.1	61.2	56.9	60.0	50.8	42.4	48.3
LoRA	76.3	61.3	51.5	51.0	51.4	52.4	33.7	21.8
$k = 0$	80.2	68.5	63.3	60.5	66.0	53.7	43.5	52.1
$k = 1$	77.1	67.5	59.7	57.5	59.3	51.4	39.5	39.9
$k = 2$	75.2	64.5	57.1	55.3	54.9	49.5	36.7	34.4
$k = 3$	74.0	63.6	55.8	54.3	54.4	47.8	34.8	32.8
$k = 4$	73.2	62.9	54.8	53.3	53.0	47.6	34.0	30.7

My Work



My Work– 2.1 CBA-Whisper (1/2)

➤ our Method – Inference of CBA-Whisper

- The proposed CBA-Whisper system consists of the following three components:
- **Preprocessing**, where we apply loudness matching, voice activity detection (VAD), and a cut-and-merge operation to segment audio into approximately 30-second input chunks;
- **Whisper large-v2 with AdaLoRA**, where we apply PEFT to adapt the model; and
- **Postprocessing**, where we combine length-based error filtering, rule-based deduplication, and neural-based normalization.

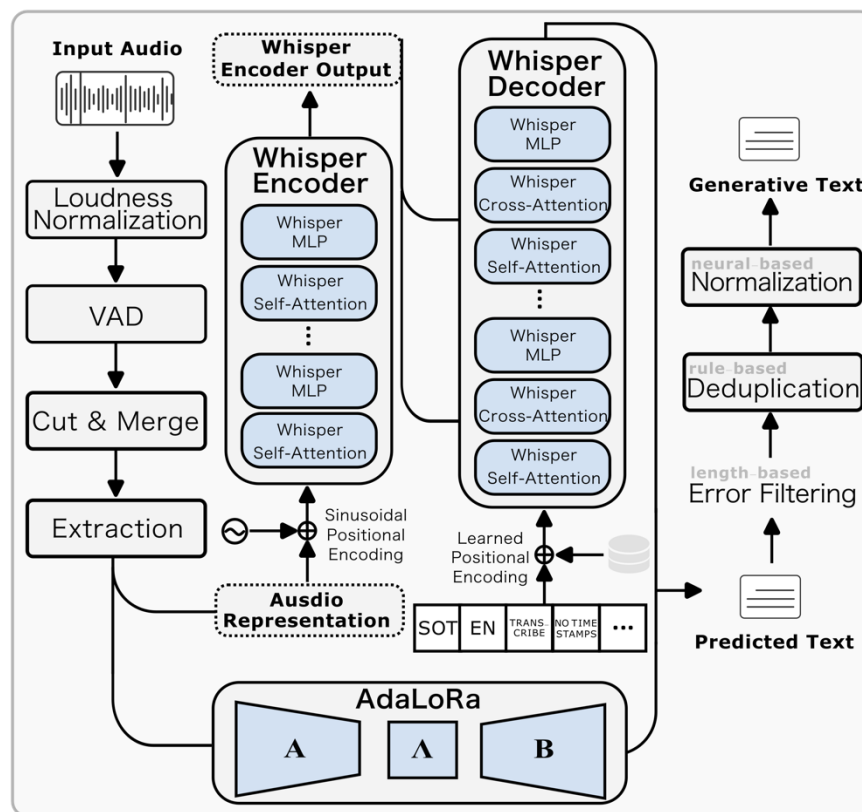


Fig. 1 High-level overview of CBA-Whisper.

My Work– 2.1 CBA-Whisper (2/2)

➤ our Method – Training of CBA-Whisper

- Our approach consists of three steps designed to optimize model performance:
- (1) We utilize Whisper large-v2 to generate initial text outputs.
- (2) **Fine-Tuning with Initial Rule:** We apply the CBA-Whisper model, trained on a selected dataset using the initial rule, as detailed in the section of Data Cleansing, to produce predicted text for new tasks.
- (3) **Fine-Tuning with Adaptive Optimization Rules:** By comparing the generative text from step (1) and (2), we identify discrepancies and refine the training dataset for the next iteration of task adaptation in step (2).

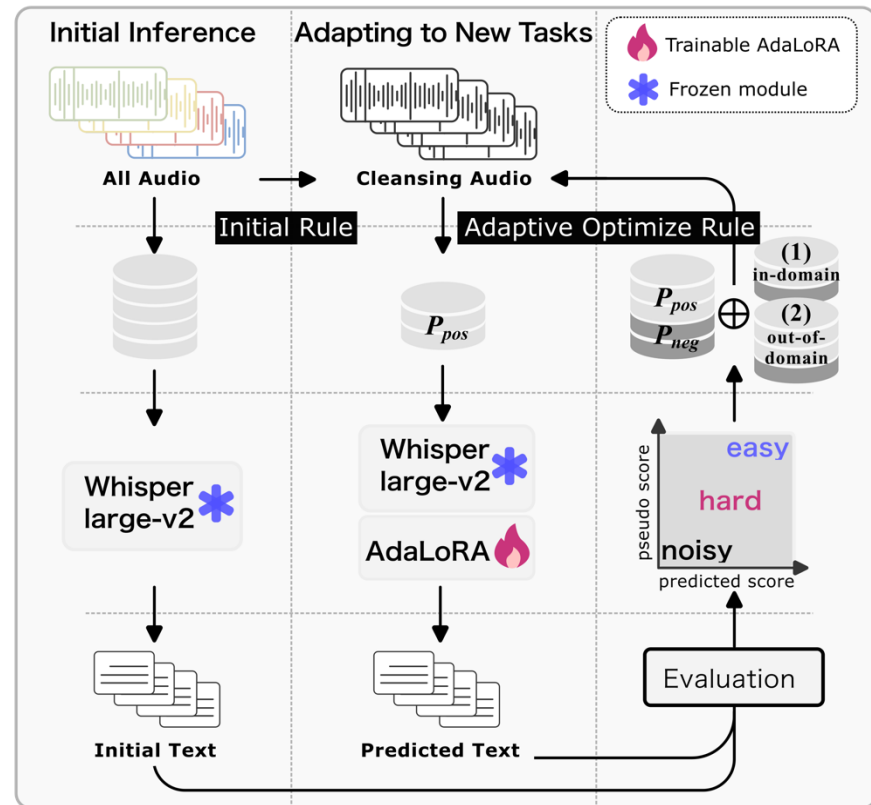


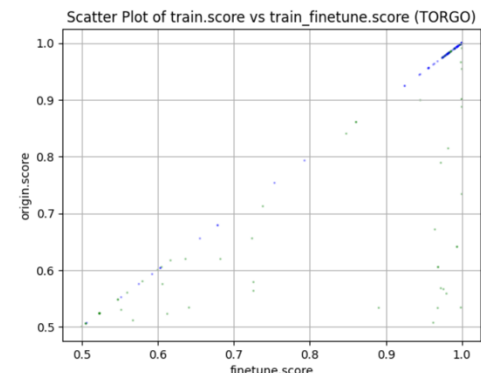
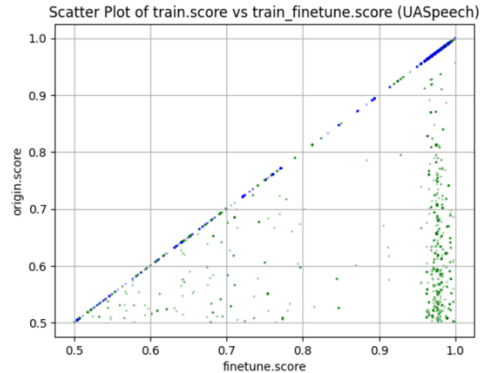
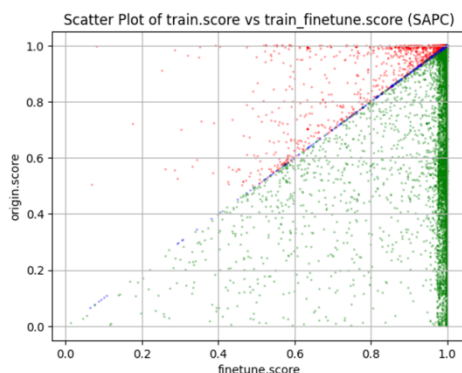
Fig. 2 Schematic diagram of CL-based AdaLoRA fine-tuning (CBA) training strategy.

My Work– 2.2 Datasets

➤ Datasets

Tab. 2 Dysarthric datasets and comparison.

Corpus	Type	Lang.	# Hour	# Speaker	# Vocab.	Available
UASpeech	Dysarthric	Eng.	102.7	29	455 words	uaspeech [✓]
TORGO	Dysarthric	Eng.	15	15	1573 words	toronto [✓]
Accessibility	Dysarthric	Eng.	771	850	1-2 weeks later	accessibility [✓]
LDRWWS	Control and Dysarthric	Ch.	7.6 and 7.5	25 and 21	10 words	lrdwvs [✓]
CDSD	Dysarthric	Ch.	124	44		CDSD [✓]



SAP Train

- 53.48-hour subset of the SPA0430 Train set
- 41.23-hour subset of the SPA0430 Dev set
- in-domain data

+

UASpeech

- 6.92-hour subset of 48.44-hour UA-Speech
- out-of-domain supplementary data

+

TORGO

- 0.13-hour subset of 2.02-hour TORGO
- out-of-domain supplementary data

We discovered an issue with data transfer that had gone unnoticed during the challenge. This resulted in the use of only 53.48 hours of 11 the SPA 2024-04-30 Train split dataset instead of the originally planned 290 hours.

My Work– 2.3 Results

➤ Results

- Our submitted system ranked **2nd** on both the official Test1 and Test2 set.
- Notably, **ngohider** had the smallest performance gap between the two test sets, while ours exhibited the largest gap.
- From the Test1 results, we observed that **single-language English models** outperform multilingual models, and **larger models** tend to yield better results.
- Ablation results organized by timeline can be observed that applying **loudness normalization** as a preprocessing step to all audio, along with our proposed **pre- and post-processing strategies**, consistently improves model performance.

Rank	Team Name	Test 1		Test2		Diff	
		WER	SemScore	WER	SemScore	WER	SemScore
1	slp-lab	5.97	92.53	8.11	88.43	+2.14	-4.10
2	witty	7.98	90.37	10.51	85.50	+2.53	-4.87
3	takagi	8.43	88.89	10.90	84.42	+2.47	-4.47
4	ngohider	10.06	86.34	11.80	83.29	+1.74	-3.05
5	SAGI	10.15	87.78	12.15	84.08	+2.00	-3.80

Model	Para. (M)	VRAM (GB)	Speed (×)	ET (sec)	Test 1	
					WER	SemScore
tiny [1]	39	~ 1	~ 10	7057	33.50	61.77
tiny.en [1]	39	~ 1	~ 10	6564	26.08	68.40
base [1]	74	~ 1	~ 7	8456	27.17	68.16
base.en [1]	74	~ 1	~ 7	7780	20.87	73.68
turbo [1]	809	~ 6	~ 8	9298	22.48	81.48
distil-large-v2 [5]	756	~ 11	~ 6	6495	15.52	79.69
distil-large-v3 [5]	756	~ 11	~ 6	6551	14.52	81.26
x-large-v2 [6]	1550	~ 10	~ 4	7100	12.18	82.50
x-large-v3 [6]	1550	~ 10	~ 4	7170	11.66	84.04

Method	Test 1		SAP-Dev		Submitted at
	WER	SemScore	WER	SemScore	
step0 finetune	10.02	87.38	9.46	90.61	2025-01-13
+deduplication	9.92	87.99	9.36	90.97	2025-01-19
+increase volumn	8.75	88.81	9.78	90.84	2025-01-24
>norm loudness	8.69	88.93	9.54	90.75	2025-01-25
all norm loudness	8.48	89.38	7.67	90.86	2025-01-27
step1 finetune	8.13	89.88	6.39	94.12	2025-01-31
+ post processing	8.09	90.17	6.45	94.07	2025-02-01
step2 finetune	7.98	90.37	6.13	94.00	2025-02-02

Conclusion and Future Work



Limitation and Conclusion

➤ Limitation

- **Trade-off:** Filtering noisy data prevents performance degradation but loses valuable text, reducing linguistic diversity.
- **Solution:** Synthesize dysarthric speech to preserve full textual distribution and augment training data.
- **Challenge:** High-quality synthetic dysarthric speech generation remains difficult (e.g., Fish-Speech results were unsatisfactory).

➤ Conclusion

- The proposed approach, based on Whisper large-v2 fine-tuned with AdaLoRA, incorporates WhisperX preprocessing, rule-based postprocessing for stuttering and hallucination, and CL with adaptive data filtering to enhance robustness and accuracy.
- This work establishes a foundational method that can be broadly applied to low-resource speech ASR fine-tuning tasks.



Thanks!